

Sensors Laboratory Experiment -3

Electronic Compass Using Hall Effect Sensors

Roll No. - [ES24f1100064](#)

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INTRODUCTION :

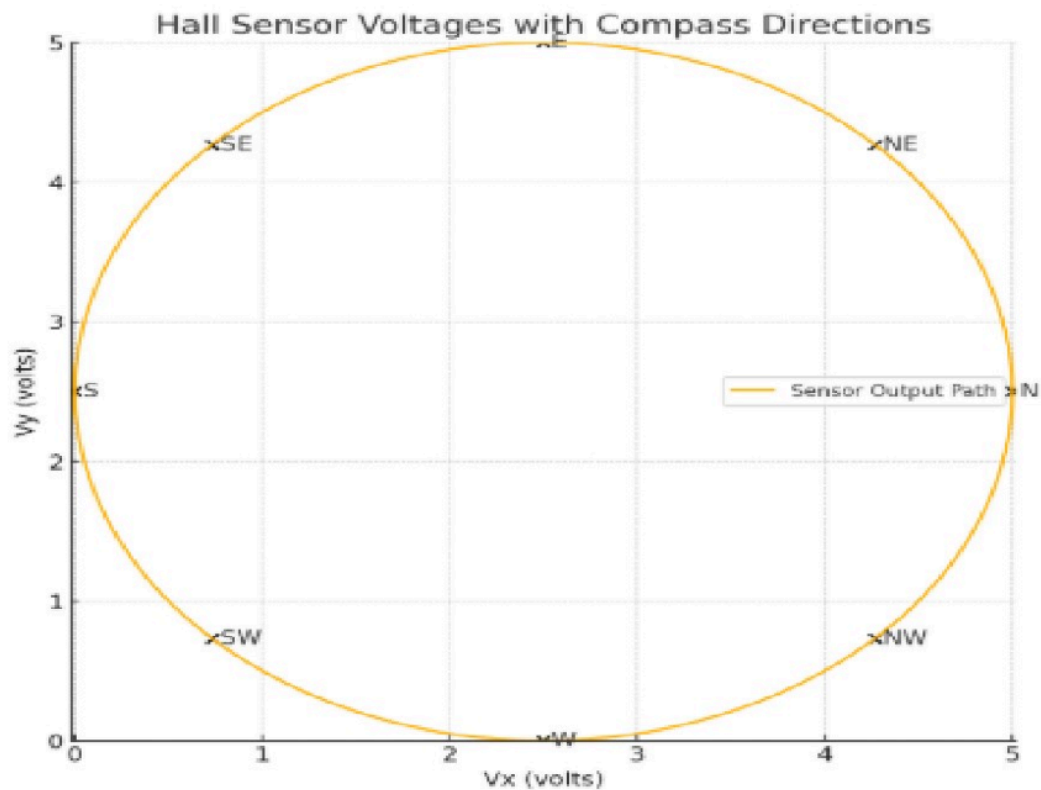
An electronic compass is a device that can be used to determine direction by sensing the Earth's magnetic field. Unlike mechanical compasses, which rely on a free-moving magnetized needle, electronic compasses use magnetic field sensors to measure the field strength along two perpendicular axes. This experiment involves designing an electronic compass using Hall Effect sensors, which generate a voltage proportional to the magnetic field strength. The compass can calculate the heading angle by processing the signals from two orthogonally placed sensors.

Hall Effect sensors, e.g. SS49 series ([datasheet](#)), operate based on the Hall Effect principle discussed in the class. The output voltage is directly proportional to the field strength, allowing magnetic field measurements. The horizontal components of the Earth's magnetic field can be determined by placing two such sensors at right angles, on a plane.

Electronic compasses are widely used in navigation systems, smartphones, drones, and robotic applications where accurate directional sensing is required. They are also employed in automotive navigation, industrial positioning, and geophysical studies. This experiment follows a design approach similar to that described in the thesis ([Marjanovic Compass](#)). The working principle of Hall Effect sensors has been covered in class, and this experiment aims to provide hands-on experience in implementing and analyzing a practical compass system.

Direction of the compass :

Suppose V_x and V_y are the voltages corresponding to the X and Y directions: As the compass rotates: $V_x = 2.5 \text{ V} + 2.5 \text{ V} \times \cos(\theta)$ $V_y = 2.5 \text{ V} + 2.5 \text{ V} \times \sin(\theta)$ Here, θ is the angle between the X-axis and magnetic North So: At 0° (North): $V_x = 5.0 \text{ V}$, $V_y = 2.5 \text{ V}$ At 90° (East): $V_x = 2.5 \text{ V}$, $V_y = 5.0 \text{ V}$ At 180° (South): $V_x = 0 \text{ V}$, $V_y = 2.5 \text{ V}$ At 270° (West): $V_x = 2.5 \text{ V}$, $V_y = 0 \text{ V}$



Design Choices Parameters of SS49E from the datasheet :

$V_{DD} = 5\text{V}$, Hall sensor idle output = 2.5V. Sensitivity: $1.4 \text{ mV/Gauss} = 14 \text{ mV/ mT}$ 0.014 mV/uT Earth's field is only about 65uT. Suppose, for design purposes we assume 300uT has the maximum input expected. Then, the output voltage = $300 * 0.014 = 4.2\text{mV} = 0.0042\text{V}$ Since the output lies in the range of 0 to 5V,

$R_g = 220$, $R = 10\text{K}$ Op amp i'm using In LT Spice AD8648 and In Hardware circuit I used MCP6004 to make the IN with 3 op amp

GAIN OF INA :

Stage 1 Gain

For a 3-op-amp INA:

$$G_1 = 1 + \frac{2R}{R_G}$$

Substitute values:

$$G_1 = 1 + \frac{2 \times 10k}{220}$$

$$G_1 = 1 + \frac{20000}{220} = 1 + 90.9 = 91.9$$

Step 2: Difference Amplifier (Stage 2)

From your circuit:

- $R_8 = R_{10} = 10k$
- $R_9 = R_5 = 10k$

So:

$$G_2 = \frac{R_{10}}{R_8} = \frac{10k}{10k} = 1$$

Final Gain

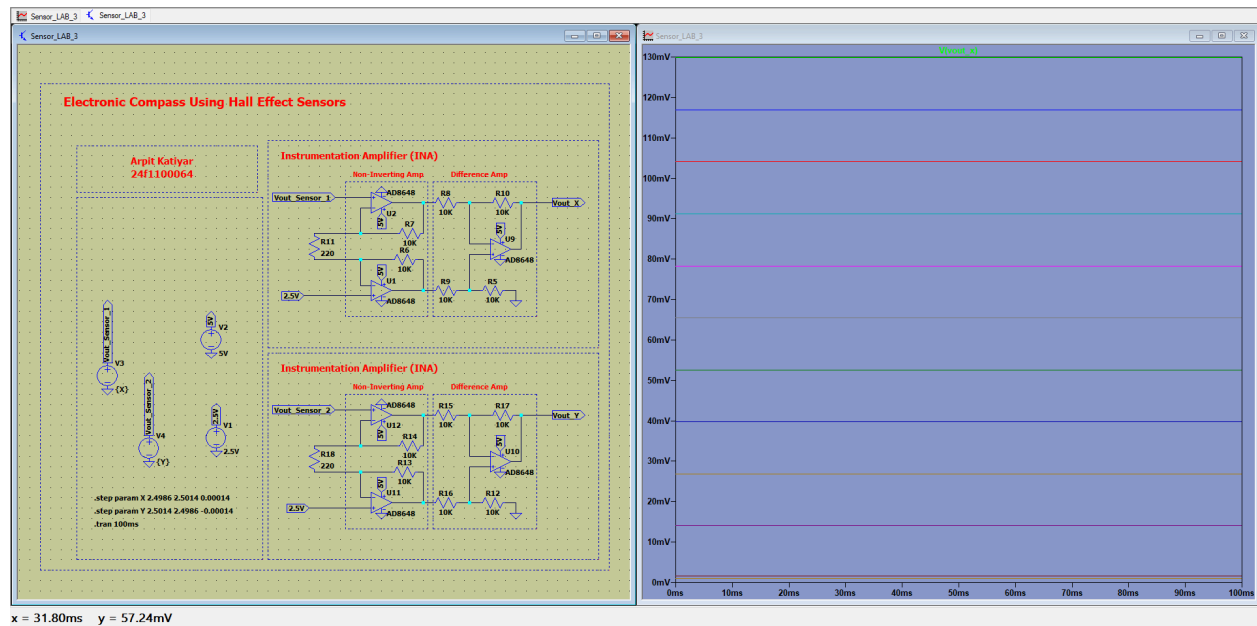
$$G_{total} = G_1 \times G_2 = 91.9 \times 1 = 91.9$$

Electronic Compass Using Hall Effect Sensors

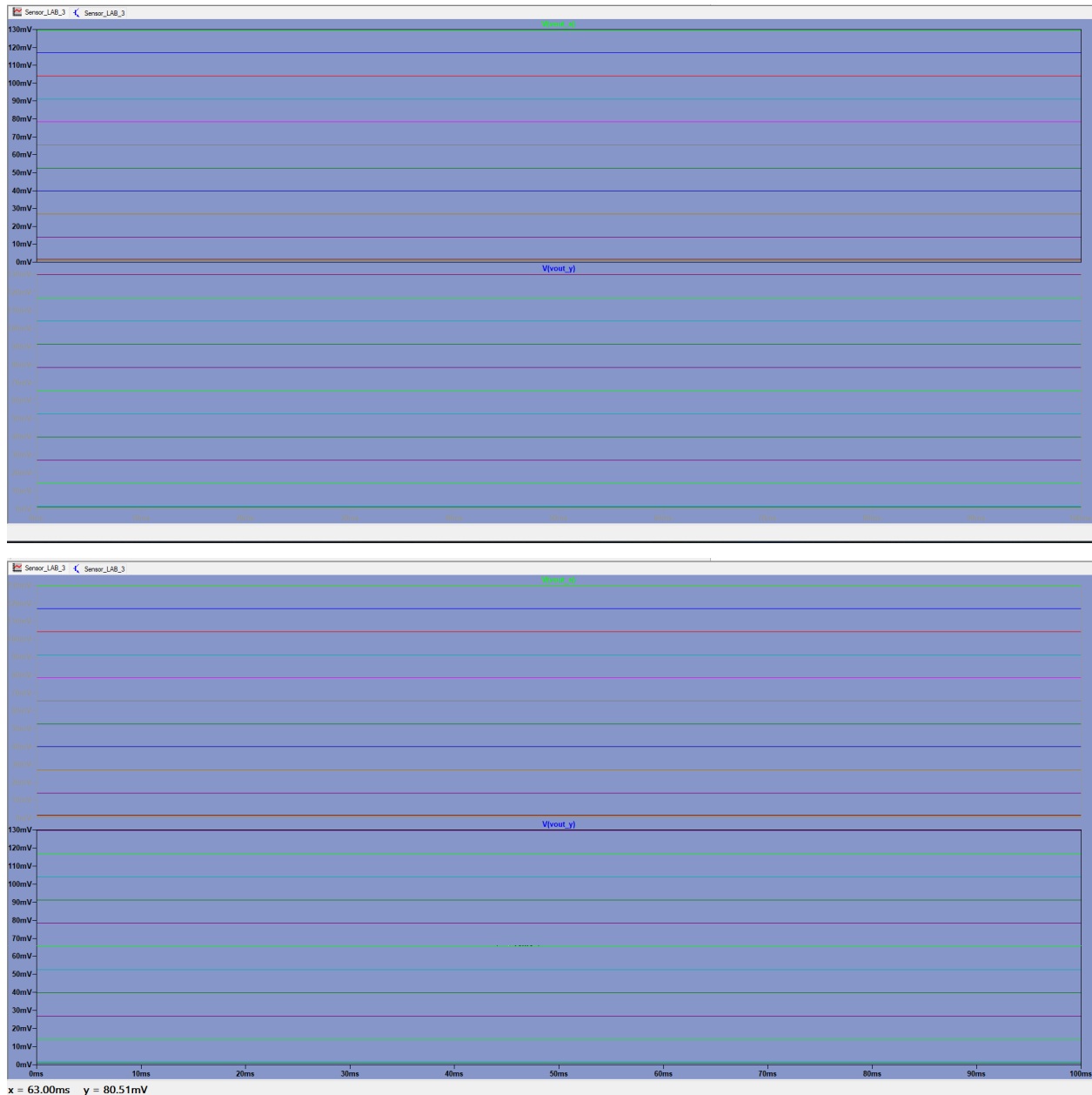
Arpit Katiyar
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Vout_Sensor_2 (Y)

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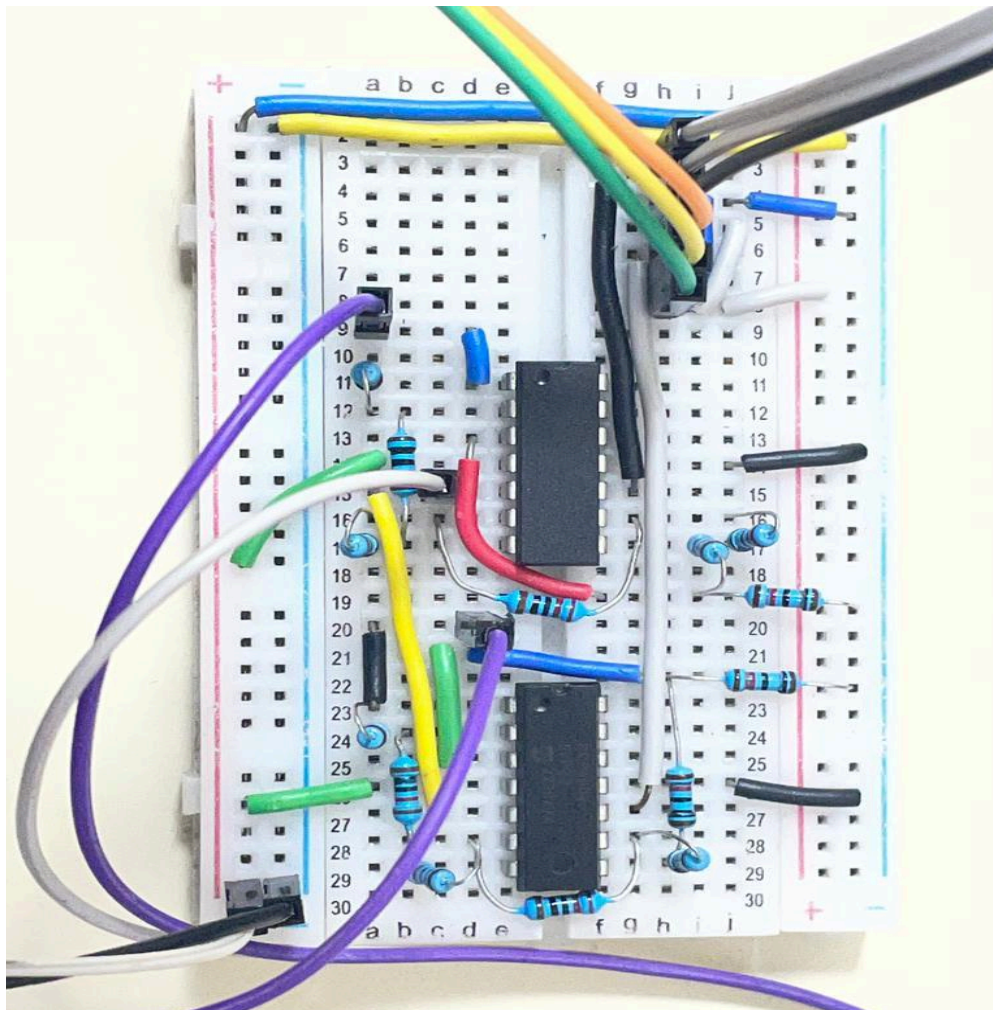
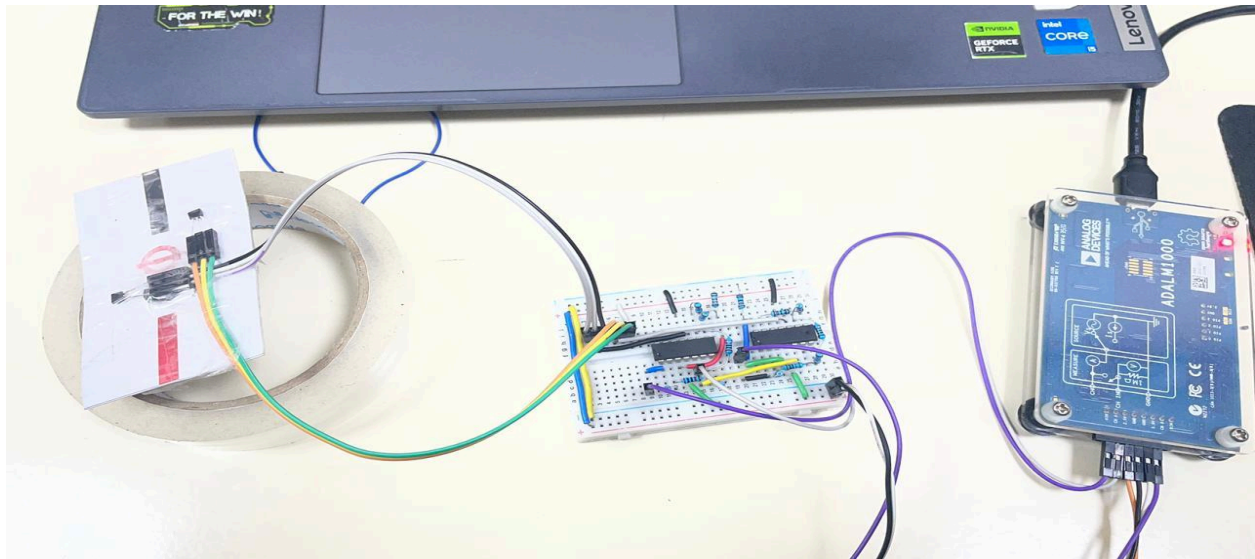


Output of the Vout_X and Vout_Y :



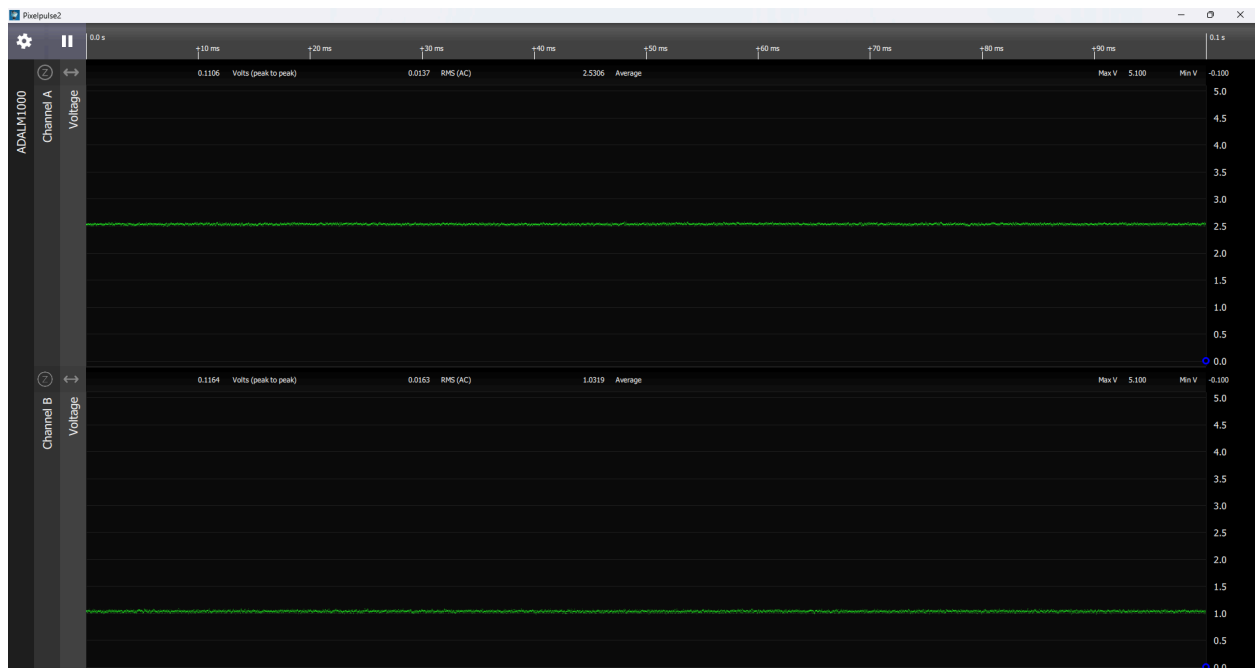
The Actual Experiment We use MCP 6004 for the Instrumentation Amplifier (INA). We use a set of resistors ($R_G = 220$ & $R = 10K$). 5V, 2.5V and GND are provided by ADALM kit. We monitor the two output voltages in Channel A and Channel B of ADALM. The following image shows the circuit on the breadboard. SS49E Stick on 90 degrees of each other on cardboard .

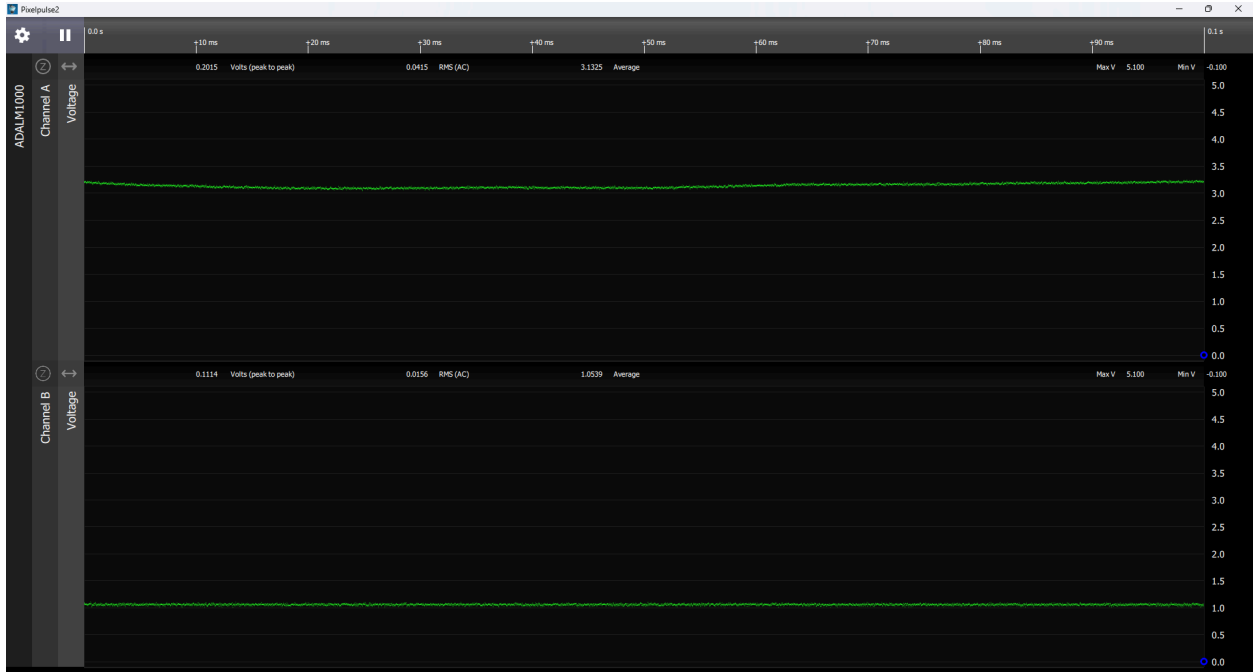
Full Breadboard Circuit :

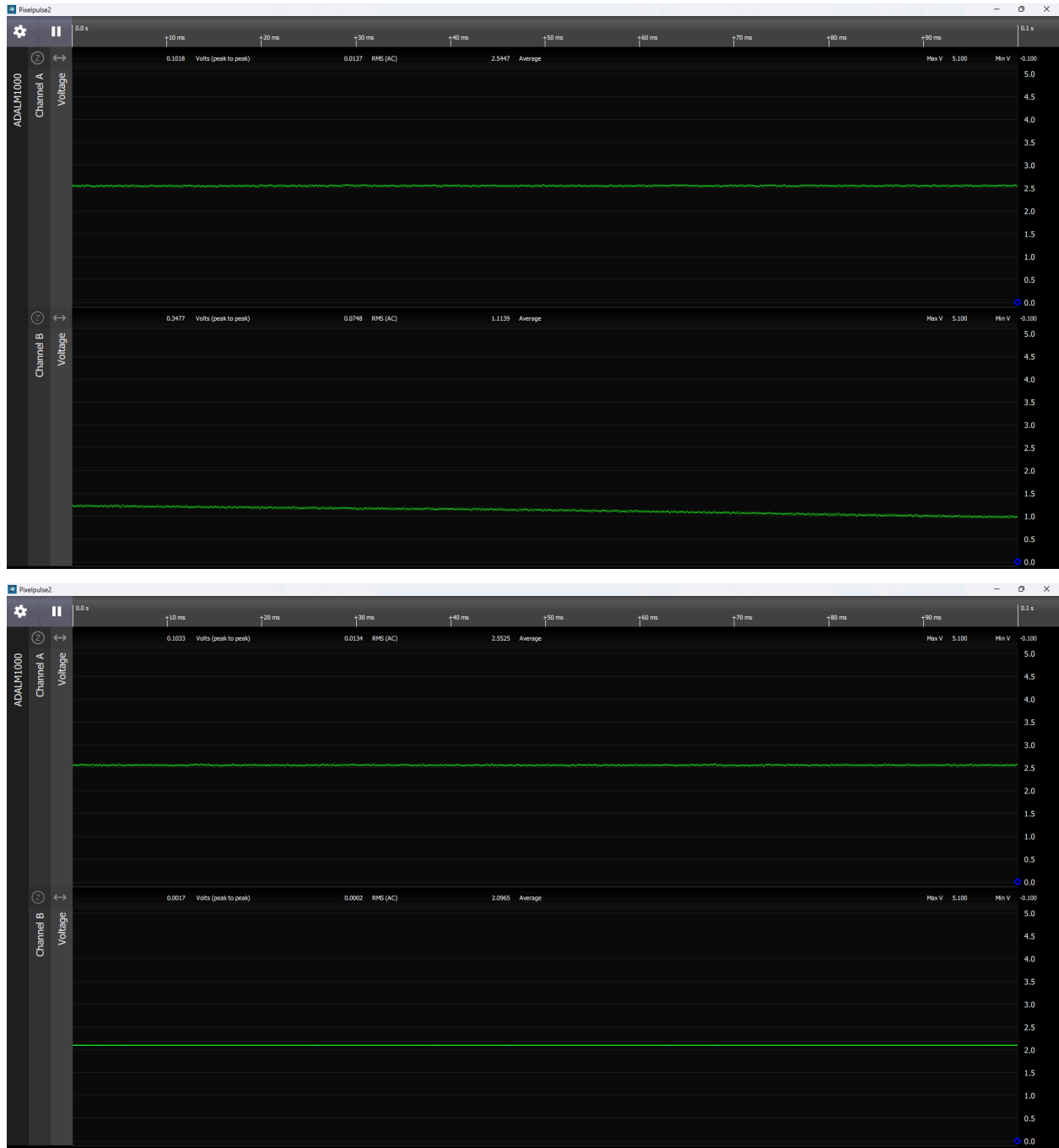




Output From the ADALM1000







Conclusion on observations :

It was quite challenging to get a stable output from this current setup. Firstly I tried a difference amplifier. But i'm not getting the proper output then, so I used Instrumentation Amplifier (INA). I just changed the gain resistor to get 91.9 gain . SS49 hall sensor output give to INA one input pin another on with 2.5V. I got the proper form output. All detailed output and circuit given above with LT Spice Simulation .